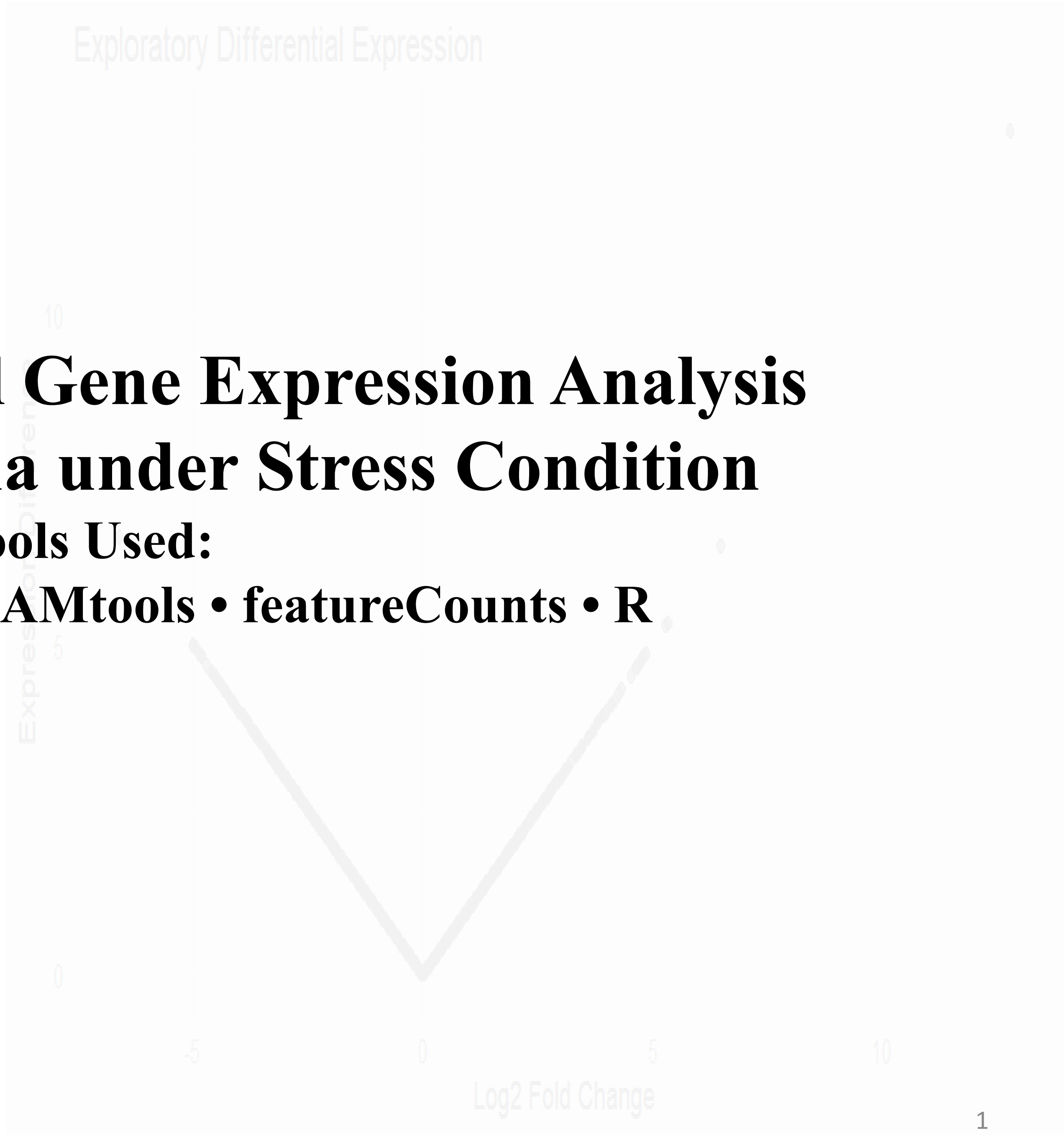


RNA-seq Differential Gene Expression Analysis
Arabidopsis thaliana under Stress Condition

Tools Used:

FastQC • HISAT2 • SAMtools • featureCounts • R

WorkFlow

FASTQ

↓

FastQC

↓

HISAT2 Alignment

↓

SAMtools

↓

featureCounts

↓

Differential Expression Analysis

↓

Visualization

RNA-seq Dataset

Organism: Arabidopsis thaliana

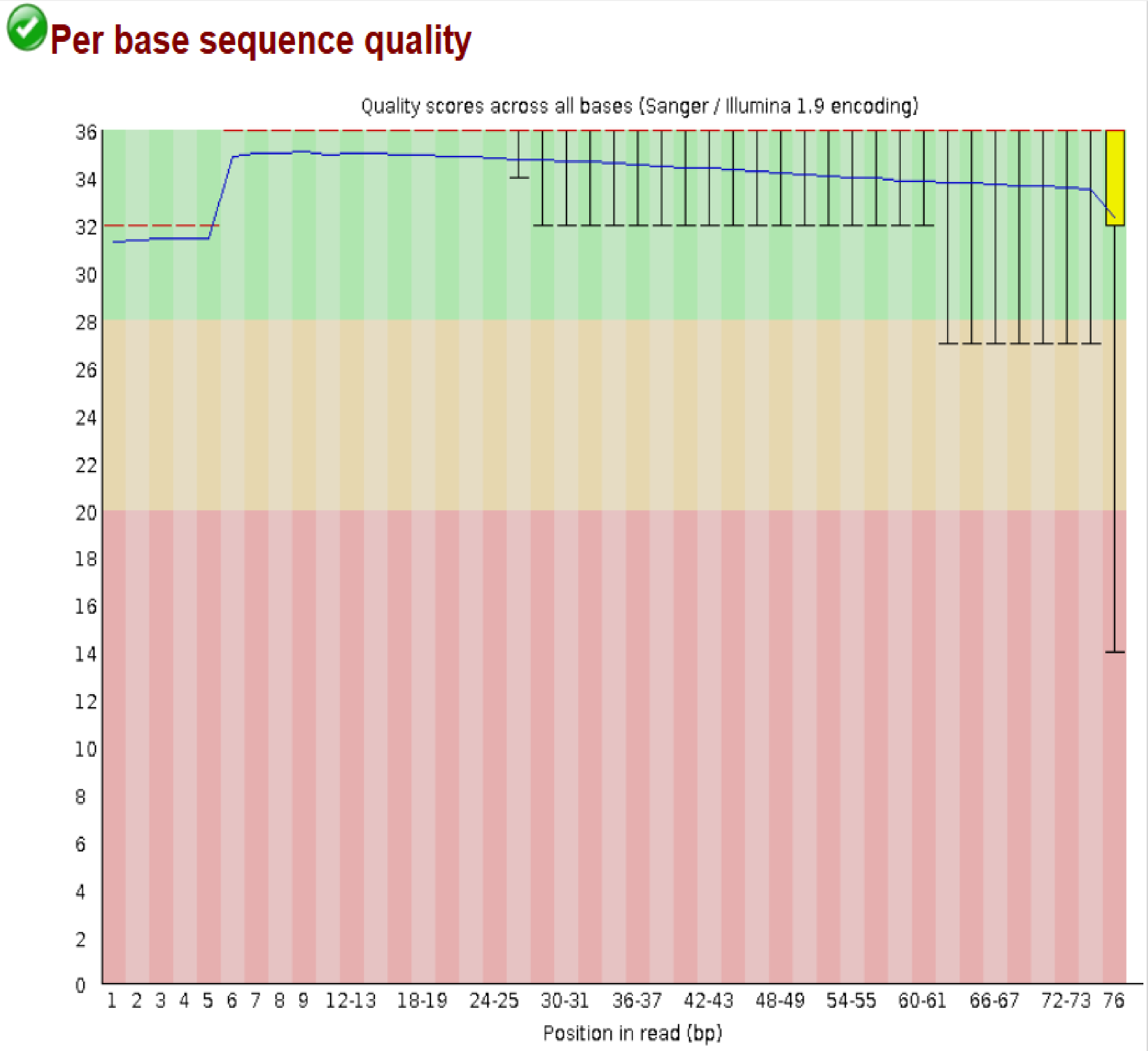
Samples:
- WT (Wild Type)
- H2B8 mutant

Sequencing Type:
- Single-end RNA-seq
- Illumina platform

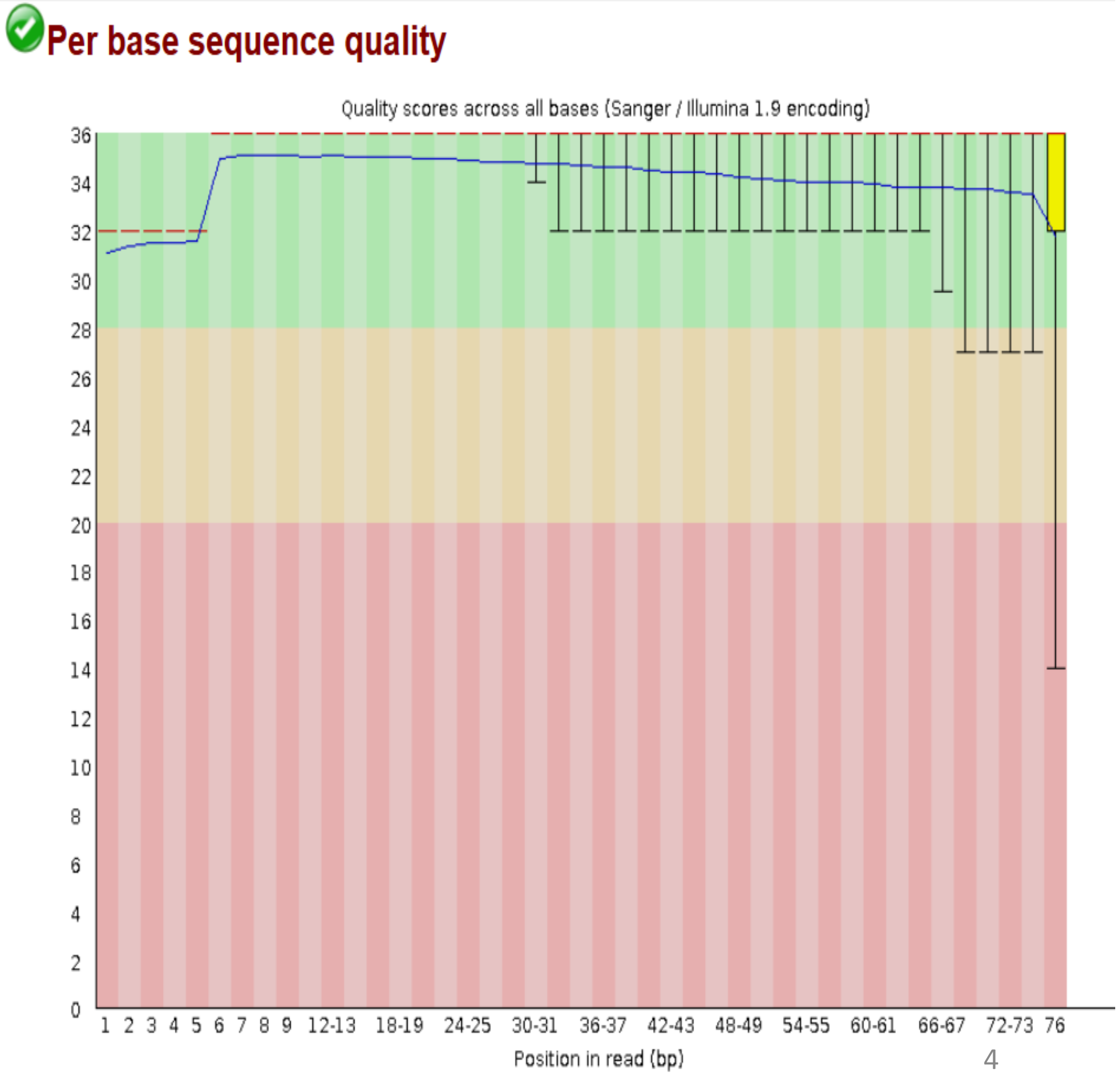
Objective:
To identify genes differentially expressed in H2B8 compared to WT.

Fastqc Report

SRR18297038

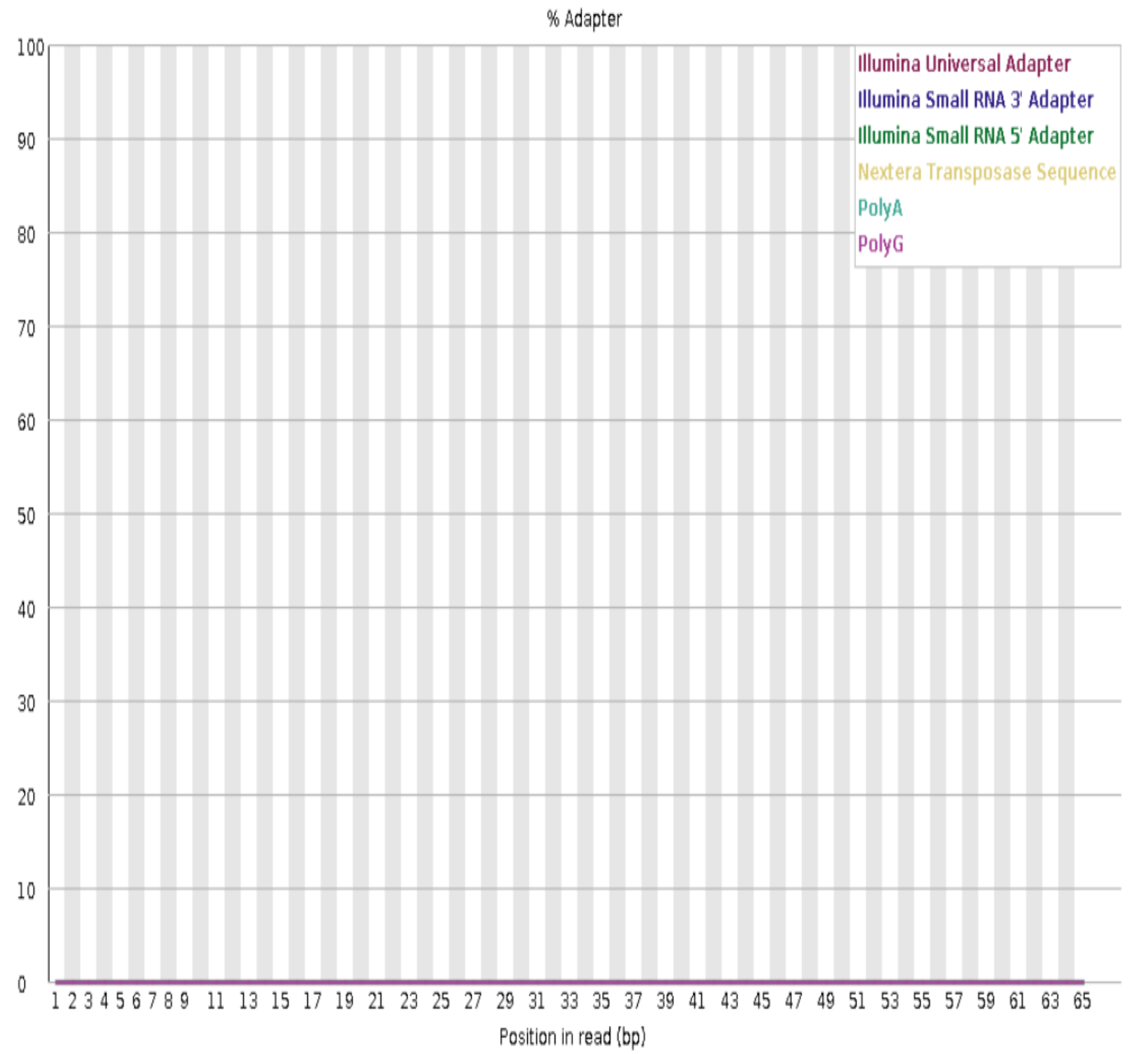


SRR18297042



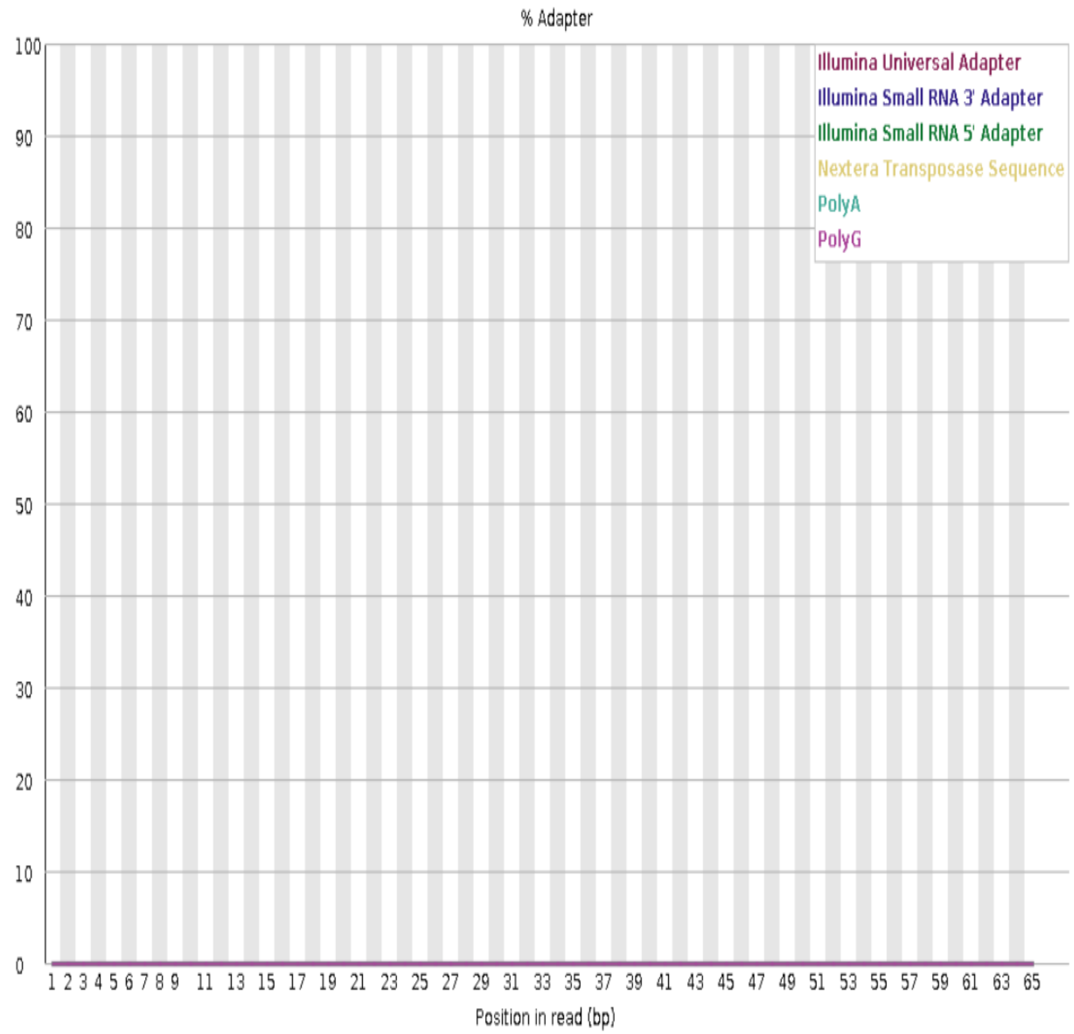
SRR18297038

Adapter Content



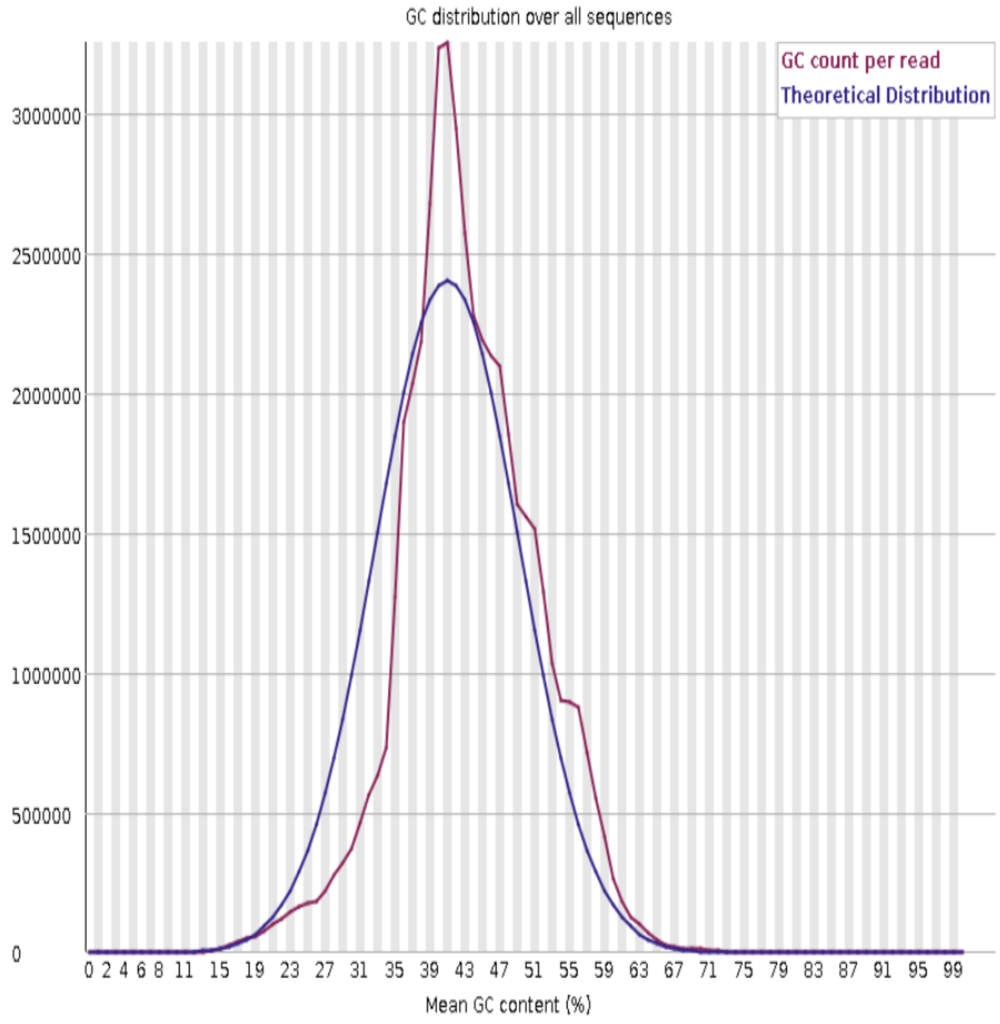
SRR18297042

Adapter Content



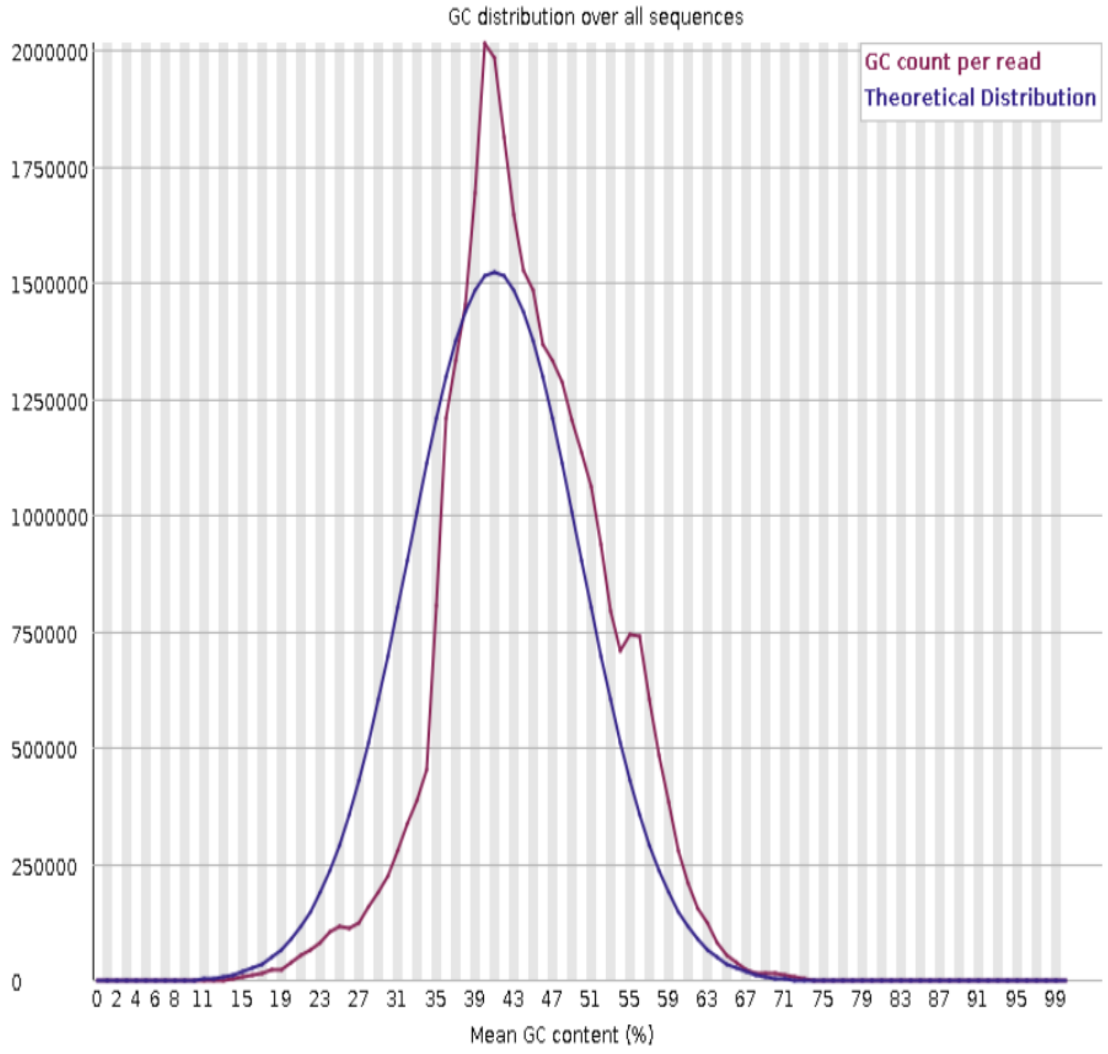
SRR18297038

Per sequence GC content



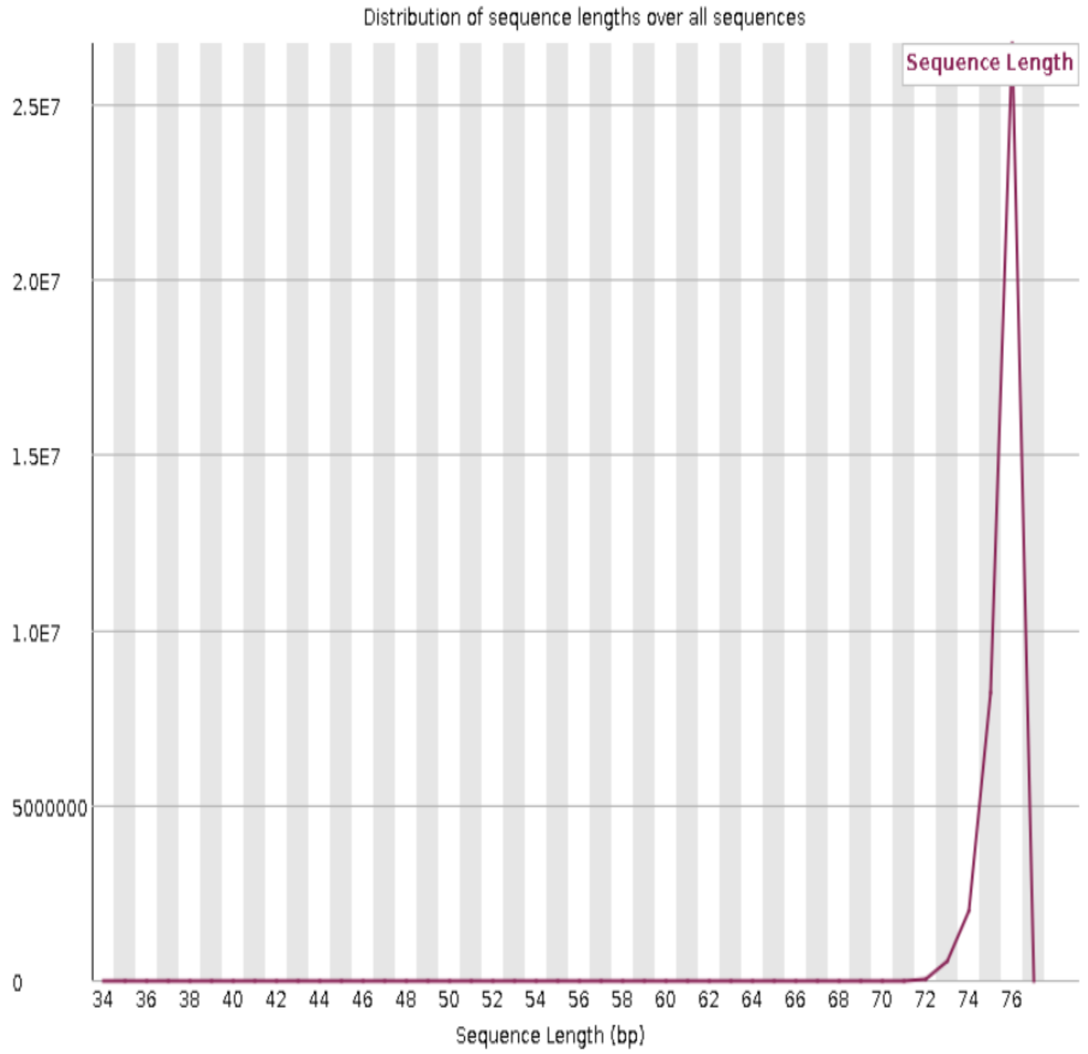
SRR18297042

Per sequence GC content



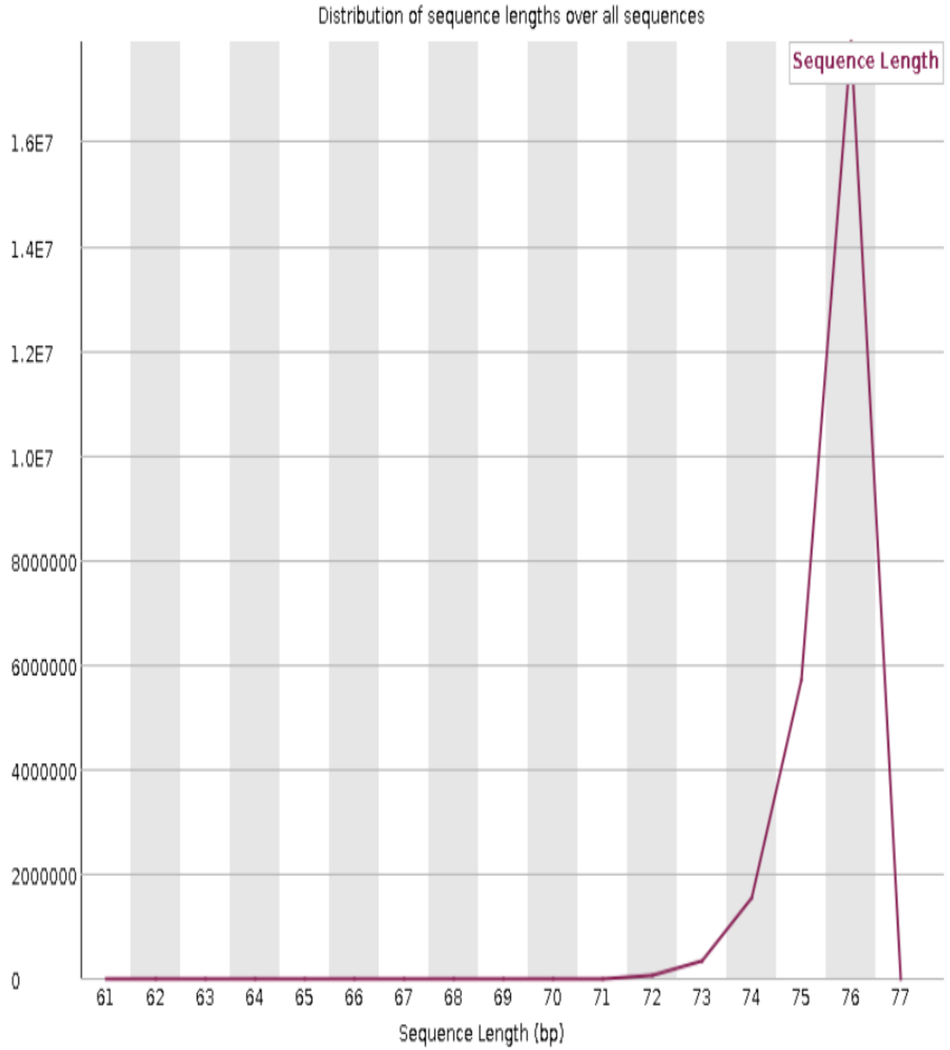
SRR18297038

Sequence Length Distribution



SRR18297042

Sequence Length Distribution



FASTQC Report Analysis

FASTQC analysis was performed to evaluate raw RNA-seq read quality before alignment.

Key Observations:
✓ High per-base sequencing quality (Q > 30 across most bases)
✓ Very low adapter contamination detected
✓ Consistent read length distribution (~76 bp)
✓ GC content showed slight deviation from theoretical distribution,
which is common in transcriptomic datasets due to biological expression bias

Conclusion:
The sequencing reads were of sufficient quality for downstream RNA-seq analysis.

Linux Workflow

godfather@Godfather: ~/Dov

```
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/data$ pwd
/home/godfather/Downloads/Unix/rnaseq_project/data
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/data$ fastqc SRR18297042_1.fastq -o /home/godfather/Dow
nloads/Unix/rnaseq_project/data
null
Started analysis of SRR18297042_1.fastq
Approx 5% complete for SRR18297042_1.fastq
Approx 10% complete for SRR18297042_1.fastq
Approx 15% complete for SRR18297042_1.fastq
Approx 20% complete for SRR18297042_1.fastq
Approx 25% complete for SRR18297042_1.fastq
Approx 30% complete for SRR18297042_1.fastq
Approx 35% complete for SRR18297042_1.fastq
Approx 40% complete for SRR18297042_1.fastq
Approx 45% complete for SRR18297042_1.fastq
Approx 50% complete for SRR18297042_1.fastq
Approx 55% complete for SRR18297042_1.fastq
Approx 60% complete for SRR18297042_1.fastq
Approx 65% complete for SRR18297042_1.fastq
Approx 70% complete for SRR18297042_1.fastq
Approx 75% complete for SRR18297042_1.fastq
Approx 80% complete for SRR18297042_1.fastq
Approx 85% complete for SRR18297042_1.fastq
Approx 90% complete for SRR18297042_1.fastq
Approx 95% complete for SRR18297042_1.fastq
Analysis complete for SRR18297042_1.fastq
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/data$ |
```

```
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/data$ fastqc SRR18297038_1.fastq -o /home/godfather/Dow
nloads/Unix/rnaseq_project/qc
null
Started analysis of SRR18297038_1.fastq
Approx 5% complete for SRR18297038_1.fastq
Approx 10% complete for SRR18297038_1.fastq
Approx 15% complete for SRR18297038_1.fastq
Approx 20% complete for SRR18297038_1.fastq
Approx 25% complete for SRR18297038_1.fastq
Approx 30% complete for SRR18297038_1.fastq
Approx 35% complete for SRR18297038_1.fastq
Approx 40% complete for SRR18297038_1.fastq
Approx 45% complete for SRR18297038_1.fastq
Approx 50% complete for SRR18297038_1.fastq
Approx 55% complete for SRR18297038_1.fastq
Approx 60% complete for SRR18297038_1.fastq
Approx 65% complete for SRR18297038_1.fastq
Approx 70% complete for SRR18297038_1.fastq
Approx 75% complete for SRR18297038_1.fastq
Approx 80% complete for SRR18297038_1.fastq
Approx 85% complete for SRR18297038_1.fastq
Approx 90% complete for SRR18297038_1.fastq
Approx 95% complete for SRR18297038_1.fastq
Analysis complete for SRR18297038_1.fastq
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/data$
```

HISAT2 Alignment Results

godfather@Godfather: ~/Dov

```
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/scripts$ ls
Align_reads.sh  Align_reads.sh4
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/scripts$ ./Align_reads.sh
Processing SRR18297038_1...
37680429 reads; of these:
  37680429 (100.00%) were unpaired; of these:
    1273349 (3.38%) aligned 0 times
    32140623 (85.30%) aligned exactly 1 time
    4266457 (11.32%) aligned >1 times
96.62% overall alignment rate
SRR18297038_1 alignment completed.
Processing SRR18297042_1...
25626959 reads; of these:
  25626959 (100.00%) were unpaired; of these:
    918015 (3.58%) aligned 0 times
    21873400 (85.35%) aligned exactly 1 time
    2835544 (11.06%) aligned >1 times
96.42% overall alignment rate
SRR18297042_1 alignment completed.
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/scripts$
```

Sorting and Indexing BAM Files for Gene Quantification

godfather@Godfather: ~/Dov

```
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/results$ samtools sort SRR18297038_1.bam -o SRR18297038_1.sorted.bam
[bam_sort_core] merging from 13 files and 1 in-memory blocks...
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/results$ ls
SRR18297038_1.bam  SRR18297038_1.sam  SRR18297038_1.sorted.bam  SRR18297042_1.bam  SRR18297042_1.sam
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/results$ samtools sort SRR18297042_1.bam -o SRR18297042_1.sorted.bam
[bam_sort_core] merging from 8 files and 1 in-memory blocks...
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/results$ ls
SRR18297038_1.bam  SRR18297038_1.sam  SRR18297038_1.sorted.bam  SRR18297042_1.bam  SRR18297042_1.sam  SRR18297042_1.sorted.bam
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/results$ samtools index SRR18297042_1.sorted.bam
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/results$ samtools index SRR18297038_1.sorted.bam
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/results$ ls
SRR18297038_1.bam  SRR18297038_1.sorted.bam      SRR18297042_1.bam  SRR18297042_1.sorted.bam
SRR18297038_1.sam  SRR18297038_1.sorted.bam.bai  SRR18297042_1.sam  SRR18297042_1.sorted.bam.bai
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/results$
```

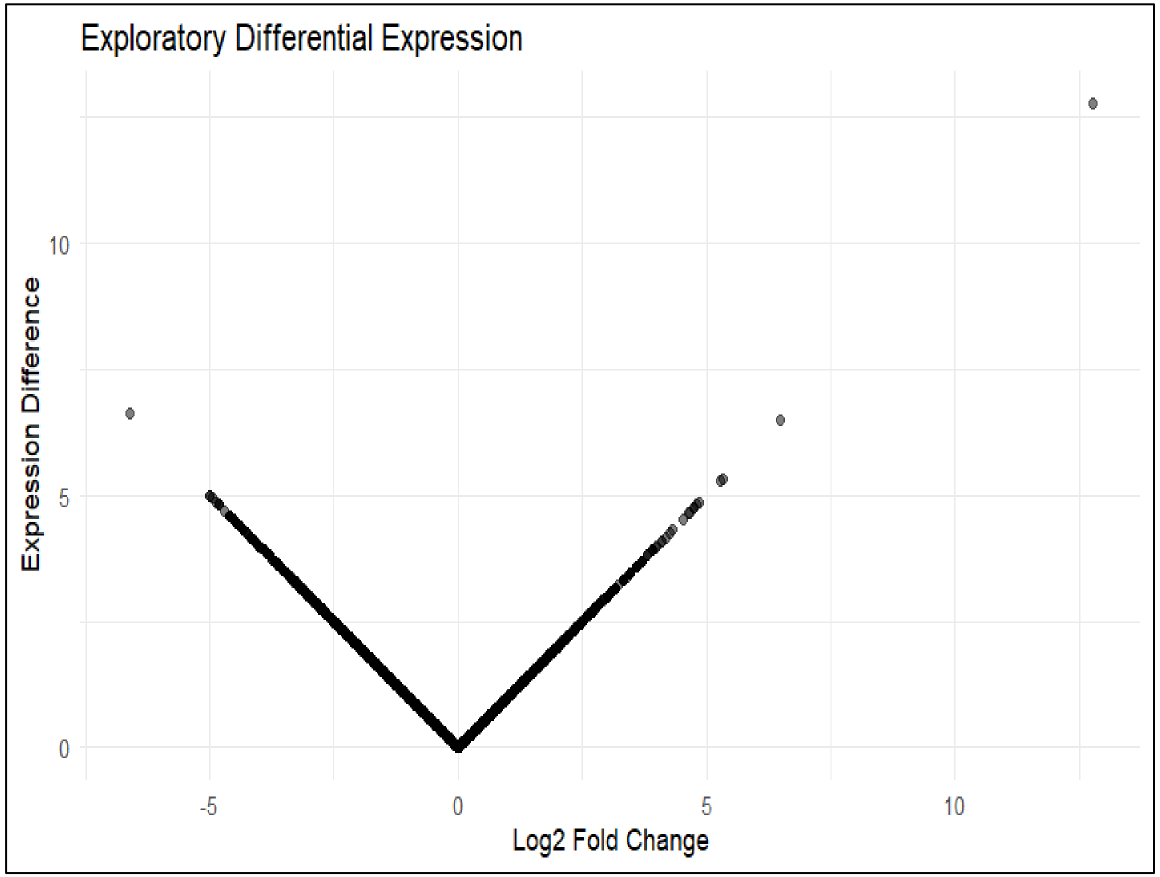
Feature Counts

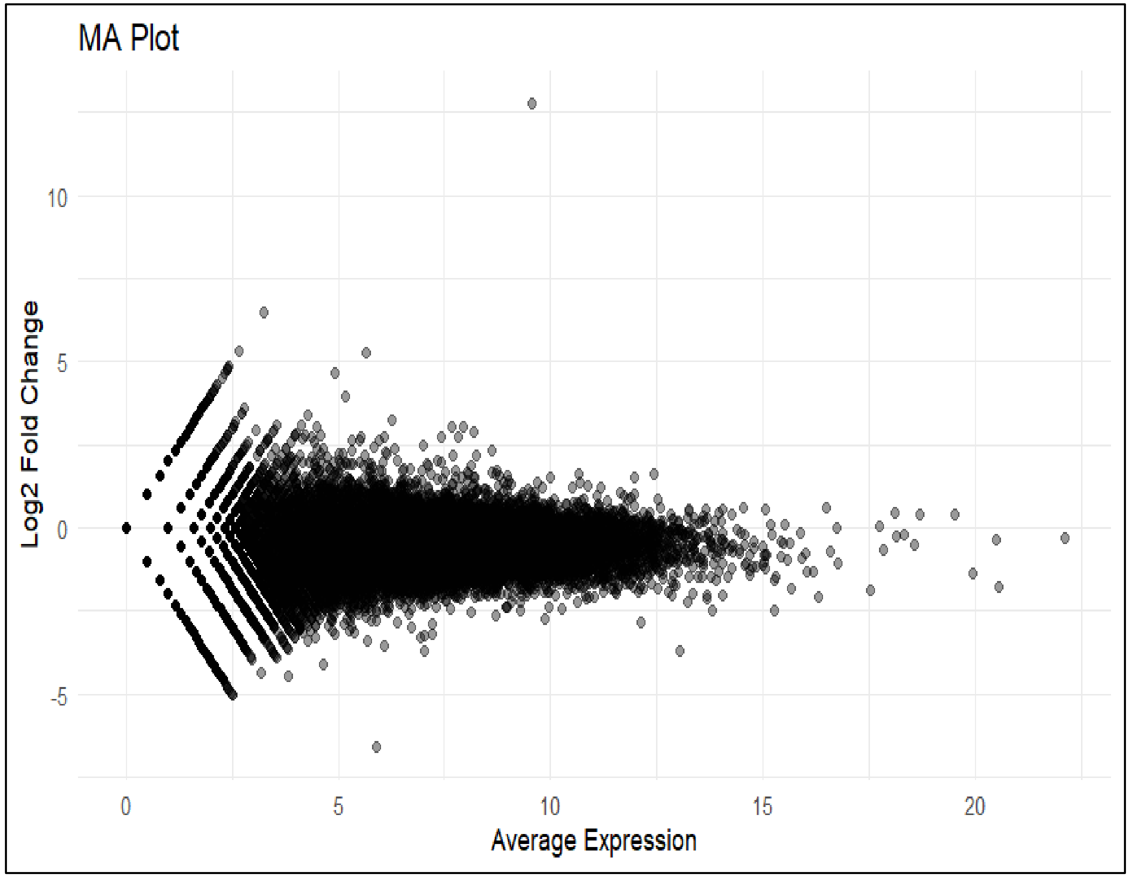
godfather@Godfather: ~/Dov

```
-bash: ./Feature_counts.sh: cannot execute: required file not found
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/scripts$ sed -i 's/\r$//' Feature_counts.sh
(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/scripts$ ./Feature_counts.sh

      v2.0.6

//========================== featureCounts setting ===========================\\
||                                                                            ||
||             Input files : 2 BAM files                                      ||
||                                                                            ||
||                           SRR18297038_1.sorted.bam                         ||
||                           SRR18297042_1.sorted.bam                         ||
||                                                                            ||
||             Output file : gene_counts.txt                                  ||
||                 Summary : gene_counts.txt.summary                          ||
||              Paired-end : no                                               ||
||        Count read pairs : no                                               ||
||              Annotation : Arabidopsis_thaliana.TAIR10.59.gtf (GTF)         ||
||      Dir for temp files : /home/godfather/Downloads/Unix/rnaseq_projec ... ||
||                                                                            ||
||                 Threads : 4                                                ||
||                   Level : meta-feature level                               ||
||      Multimapping reads : not counted                                      ||
|| Multi-overlapping reads : not counted                                      ||
||   Min overlapping bases : 1                                                ||
||                                                                            ||
\\============================================================================//

//================================= Running ==================================\\
||                                                                            ||
|| Load annotation file Arabidopsis_thaliana.TAIR10.59.gtf ...                ||
||    Features : 313952                                                       ||
||    Meta-features : 32833                                                   ||
||    Chromosomes/contigs : 7                                                 ||
||                                                                            ||
|| Process BAM file SRR18297038_1.sorted.bam...                               ||
||    Single-end reads are included.                                          ||
||    Total alignments : 42417021                                             ||
||    Successfully assigned alignments : 29079690 (68.6%)                     ||
||    Running time : 0.22 minutes                                             ||
||                                                                            ||
|| Process BAM file SRR18297042_1.sorted.bam...                               ||
||    Single-end reads are included.                                          ||
||    Total alignments : 28811913                                             ||
||    Successfully assigned alignments : 19808188 (68.7%)                     ||
||    Running time : 0.17 minutes                                             ||
||                                                                            ||
|| Write the final count table.                                               ||
|| Write the read assignment summary.                                         ||
||                                                                            ||
|| Summary of counting results can be found in file "/home/godfather/Downloa  ||
|| ds/Unix/rnaseq_project/results/gene_counts.txt.summary"                    ||
||                                                                            ||
\\============================================================================//

(rnaseq_env) godfather@Godfather:~/Downloads/Unix/rnaseq_project/scripts$ |
```

Differential Expression Visualization in R

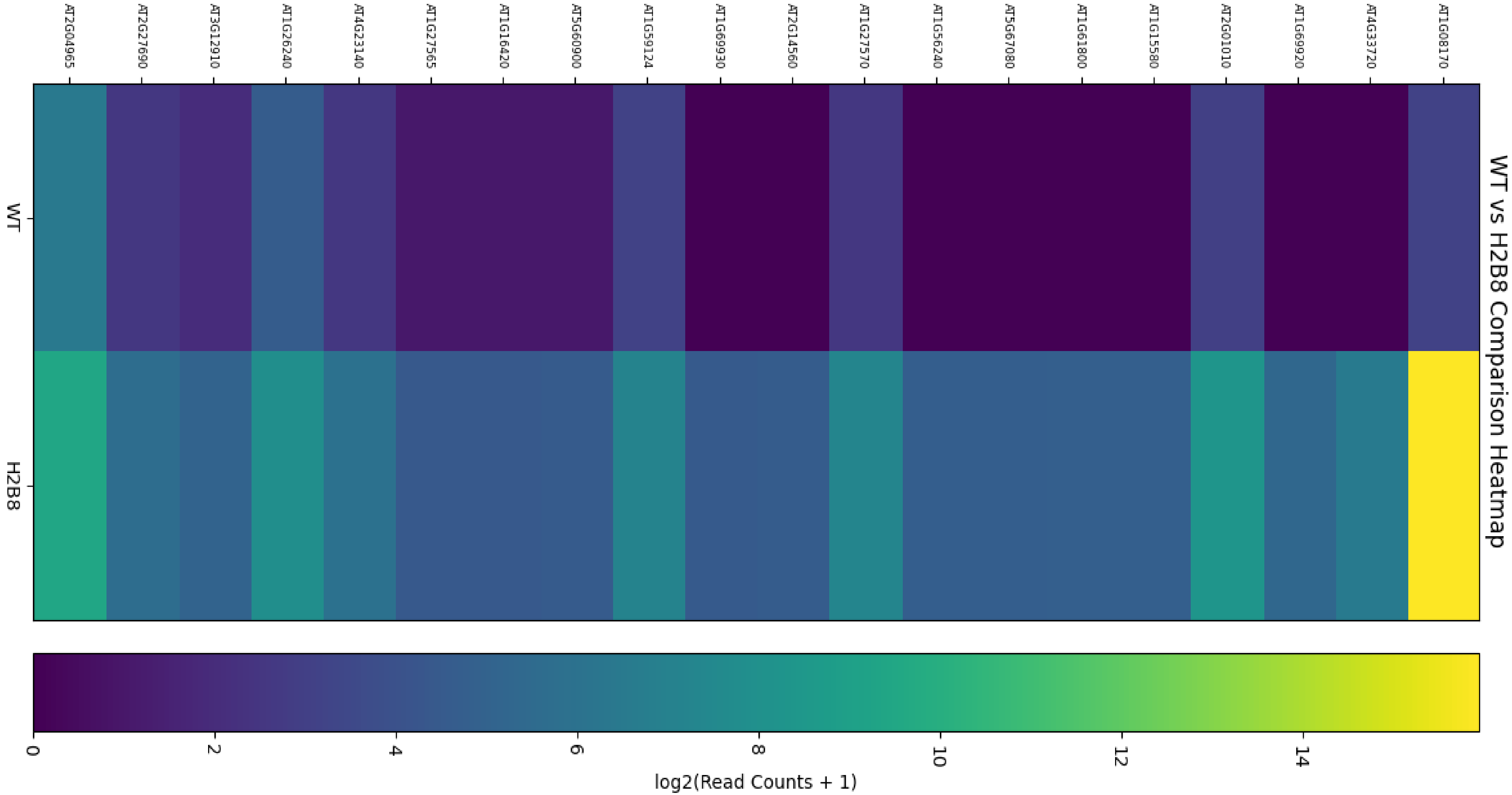
Volcano Plot + MA Plot





Differential expression analysis revealed significant transcriptional changes.

Heatmap



Top differentially expressed genes clearly separate WT and H2B8 samples.

Biological Interpretation

AT1G08170 showed strong upregulation in H2B8 samples, suggesting possible involvement in stress-responsive pathways.

Complete RNA-seq pipeline implemented using Linux, Bash, R, and NGS tools.

Complete RNA-seq workflow and analysis scripts available on GitHub.